Joseph Crockett

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Profile

Final-year Computer Science with AI student at the University of York who has delivered Production software for over 16 months at Cassini Systems, a London based Fintech. Shipped Python features used by institutional clients including BlackRock, took an AWS service from design to deployment, and resolved penetration-test findings so the platform met client security requirements.

Technical Skills

- **Languages:** C#, Python, TypeScript, JavaScript, SQL, Bash, HTML/CSS
- **Backend & frontend:** .NET, FastAPI, Node.js, React, Svelte, Vite
- **Cloud & infrastructure:** AWS, Docker, Docker Compose, Proxmox VE, Linux server administration
- **Networking & security:** Tailscale, WireGuard, Nginx Proxy Manager (reverse proxy, TLS), Pi-hole DNS, vulnerability remediation
- **Tooling:** Git, GitHub, GitLab, CI/CD, REST APIs, Postman, Claude Code
- **Spoken languages:** English (native), Japanese (conversational)

Work Experience

Software Development Intern · Cassini Systems (fintech) · London / remote · Jan 2024 – present

7-month full-time placement, then 16 hours per week alongside university.

- Delivered an event-driven Microsoft Teams chatbot end to end, designing, building and deploying it on AWS to give the team real-time developer workflow statistics from Jira and GitLab.
- Shipped Python backend services and full-stack features to Production on financial analytics platforms used by institutional clients including BlackRock.
- Designed MySQL schemas and wrote the queries and data-processing logic behind backend services.
- Built and shipped web interfaces and internal developer tools for the engineering team in TypeScript, React and Svelte (Vite).
- Resolved security vulnerabilities raised in formal penetration testing, enabling the platform to meet client security requirements.
- Met agreed deadlines through term time while balancing the role with a full university workload.

Certifications

- **Anthropic**: Building with the Claude API; Introduction to MCP; MCP Advanced Topics; Claude Code in Action; Introduction to Sub-Agents; AI Fluency: Framework & Foundations
- **Harvard University:** CS50x (Introduction to Computer Science); CS50P (Introduction to Programming with Python)

Projects

Full-stack web applications, each designed and shipped end to end using AI-assisted development and published on GitHub.

strata · C#, .NET, React

- Built a drag-and-drop website builder with a section-based page editor on a .NET backend and React frontend, to learn the patterns behind modern website platforms.

tally · C#, .NET, React

- Shipped a calorie-intake tracker with barcode scanning, integrating UK food databases as external data sources for food lookup.

train-hard · Python (FastAPI), Svelte

- Built and shipped a gym tracker that logs exercises, compares progress over time and tracks body-weight changes.

Self-Hosted Server Infrastructure · Proxmox VE, Linux, Docker Compose, Tailscale, WireGuard, Nginx Proxy Manager, Pi-hole

- Built and ran a three-node Proxmox VE cluster, provisioning and managing many VMs and LXC containers across a range of operating systems.
- Operate Linux servers hosting more than ten containerised services and automated data pipelines, defined in Docker Compose and version-controlled in Git for repeatable deployments.
- Secured remote access through a Tailscale mesh VPN, with Nginx Proxy Manager for domain routing, reverse proxying and automatic TLS renewal, plus Pi-hole for network-wide DNS filtering.
- Diagnosed and fixed a kill-switch firewall rule that blocked traffic between containers behind a shared WireGuard VPN container (gluetun), restoring service by allowing the internal Docker subnet.

Education

BSc (Hons) Computer Science with Artificial Intelligence · University of York · Sept 2024 – Jun 2027 (expected)

- Final year, on track for a First.
- Final-year project: *LLM-Assisted Quality Review of Agile User Stories*, a Jira-integrated tool that reviews user stories for quality using rule-based checks and a pretrained LLM, building on Jira integration work from Cassini.
- Relevant modules: Artificial Intelligence, Data Structures & Algorithms, Software Engineering, Database Systems, Distributed Systems, Computer Architecture.

Academic Year Abroad · Ritsumeikan University, Osaka, Japan · Sept 2025 – Aug 2026

- One-year academic exchange studying Computer Science alongside Japanese language studies.

A-Levels · St Mary's Sixth Form, Bishop's Stortford · 2023

- Computer Science (A), Mathematics (A), Economics (B).